

Assignment: Medicaid Lottery and Causal Inference

Dataset

Use the **Oregon Health Insurance Experiment (OHIE)** dataset: `OHIEresults.csv`

Objective

Analyze whether **randomized Medicaid lottery selection affects healthcare utilization** and understand the role of randomization and Instrumental Variables (IV) in causal inference.

1. Data Exploration

Using Python:

- Load the dataset.
- Report the dataset shape.
- Display the first 10 observations.
- Check for missing values.
- Calculate the number and percentage of lottery-selected and non-selected individuals.

2. Treatment Effect

Using `selected` as the treatment variable and the appropriate healthcare-utilization variable:

- Calculate the mean outcome for the selected group.
- Calculate the mean outcome for the non-selected group.
- Calculate the Average Treatment Effect (ATE):

$$ATE = E[Y_{\text{selected}} - Y_{\text{not selected}}]$$

Interpret the estimated ATE in **2–3 sentences**.

3. Visualization

Create a **bar chart** comparing the average healthcare utilization of:

- Lottery-selected individuals
- Non-selected individuals

Include an appropriate **title, x-axis label, and y-axis label**.

4. Regression and Hypothesis Testing

Estimate the following regression model:

$$Y_i = \beta_0 + \gamma \text{Selected}_i + \varepsilon_i$$

Report:

- Treatment coefficient (γ)
- Standard error
- t-statistic
- p-value

Test the following hypotheses:

$$H_0: \gamma = 0$$

$$H_1: \gamma \neq 0$$

Use a **5% significance level** ($\alpha = 0.05$). State whether you **reject or fail to reject** H_0 and briefly interpret the result.

5. Compliance and Instrumental Variables

Answer the following questions briefly:

- 1 Why is **lottery selection** different from **actual Medicaid enrollment**?
- 2 Why can comparing **Medicaid enrollees and non-enrollees** produce a biased estimate of the treatment effect?
- 3 Why can **lottery selection** be used as an **instrumental variable** for actual Medicaid enrollment?